Amir Mohseni

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Education

MSc Advanced Computer Science Oct 2026 – Present
University of Oxford, Oxford, United Kingdom

- Focus on **Artificial Intelligence**; expected completion September 2027.

Bachelor of Computer Science Sep 2023 – Jun 2026
Maastricht University, Maastricht, The Netherlands

- Graduated **with distinction (*cum laude*)**; **GPA:** 8.9 / 10.0.
- Selected for the **MARBLE Research Program (Honours Track)**.
- **Thesis:** *Uncertainty-Aware Legal Query Classification with Small Language Models*.
- **Supervisor:** Gerasimos Spanakis.

Publications

CurveBench: A Benchmark for Exact Topological Reasoning over Nested Jordan Curves
Amirreza Mohseni, Mona Mohammadi, Morteza Saghafian, and Naser Talebizadeh Saradari.
NeurIPS 2026 *(accepted, poster)* | arXiv:2605.14068

Legal Guidance in the Wild: How Users Seek Legal Help in Real-World LLM Conversations
Amirreza Mohseni and Gerasimos Spanakis.
JURIX 2026 *(accepted)*

Experience

Student Researcher, Turing – Remote Apr 2025 – Present

- Leading a team of student researchers focused on **LLM modeling, reinforcement learning, and evaluation**.
- Fine-tuning in-house models using reinforcement learning to improve **tool use, code generation, and reasoning**.
- Trained **reward models** on EduArena preference data, improving human-preference prediction accuracy from **62% to 75%**.
- Presented the **EduArena** research platform at **ICML 2025**.

Generative AI Researcher, Scale AI – Remote Sep 2024 – Apr 2025

- Collaborated with ML researchers and PhD students to fine-tune and benchmark **Gemma 2** and **Llama 3**.
- Improved reasoning on **Math-500, MMMU, and GPQA Diamond** using SFT and DPO methods inspired by **STaR** and **V-STaR**.
- Delivered executive reports to enterprise clients summarizing experimental results and model evaluations.

Research Assistant, Maastricht University – Maastricht, The Netherlands Jun 2024 – Sep 2024

- Collected and cleaned large-scale social-media data from company Facebook pages using a licensed API.
- Trained and evaluated regression models to analyze engagement and social trends in response to news events.

Selected Research and Projects

Bachelor's Thesis Research: RLVR, Overconfidence, and Abstention

- Investigated **overconfidence in RLVR-trained LLMs** for legal-query routing and how **abstention** could mitigate it by restoring the models' ability to express uncertainty.
- Compared encoder classifiers and small LLMs trained with supervised fine-tuning and RLVR; improved legal-guidance agreement with frontier-model reference labels from **75.7% to 91.2%** using a 4B-parameter model.

Reasoning Router Blog / Models

- Developed a router to select *thinking* or *non-thinking* inference, reducing token usage by nearly **50%** while maintaining performance within **5%** of the best model's accuracy.

Achievements

- **ICPC World Finalist** (45th Annual, 2022). ICPC Profile
- **Bronze Medalist**, ACM-ICPC Regional Contest (2023).
- Founded the **ICPC Club at Maastricht University**; organized weekly training sessions and contests, and qualified for **NWERC** in all eligible years.
- **Ranked 482nd** among approximately 120,000 mathematics participants (**Top 0.4%**) in Iran's National University Entrance Exam (2021).
- Reached the **Top 120 nationally** out of **10,000 first-round entrants** (**Top 1.2%**) in Iran's National Olympiad of Informatics (2020).

Skills

Programming Languages: Python, C/C++, Java

Machine Learning & AI: PyTorch, NumPy, vLLM, Hugging Face TRL

Databases & Cloud: SQL, GCP (Vertex AI)